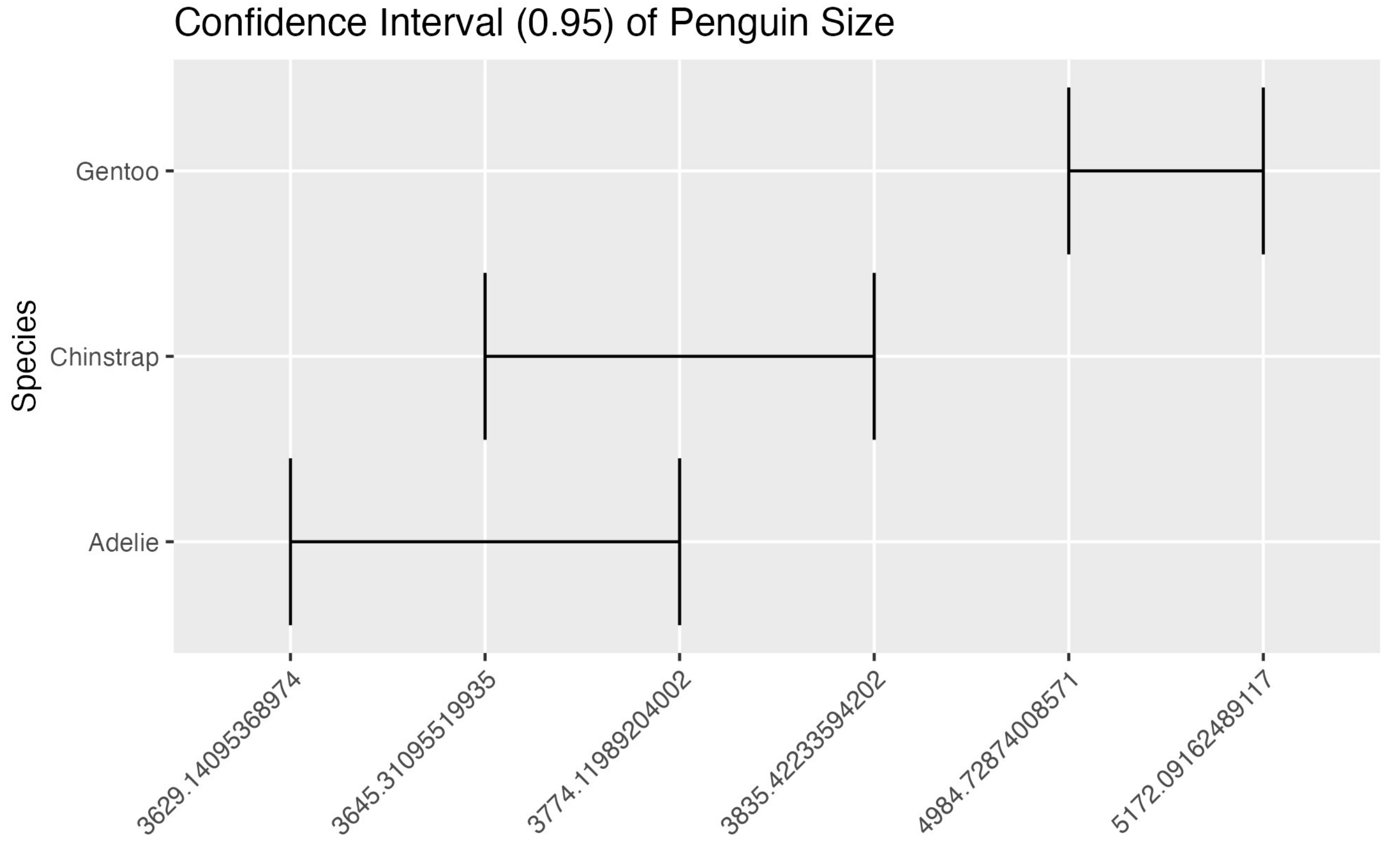
Hypothesis Testing Overview

Exercise

The plot below shows the 95 percent confidence interval of body mass of different species of penguins. How do we feel about the following statement:

Based on our data, we can conclude that Chinstrap penguins have higher body mass on average than Adelie penguins.



a) Strongly Agree

b) Agree

c) Neutral

d) Disagree

e) Strongly Disagree

Hypothesis Testing

A systematic way to **make decisions** about the conclusions we can draw from our data.

We have a preconceived notion about the world.

We are presented with an alternative view.

We review the evidence.

We determine if the evidence is enough.

If the evidence is sufficient, we reject our preconceived notion and accept the alternative view as our new reality.

Continue…

We have a preconceived notion about the world.
All cats are meanies.

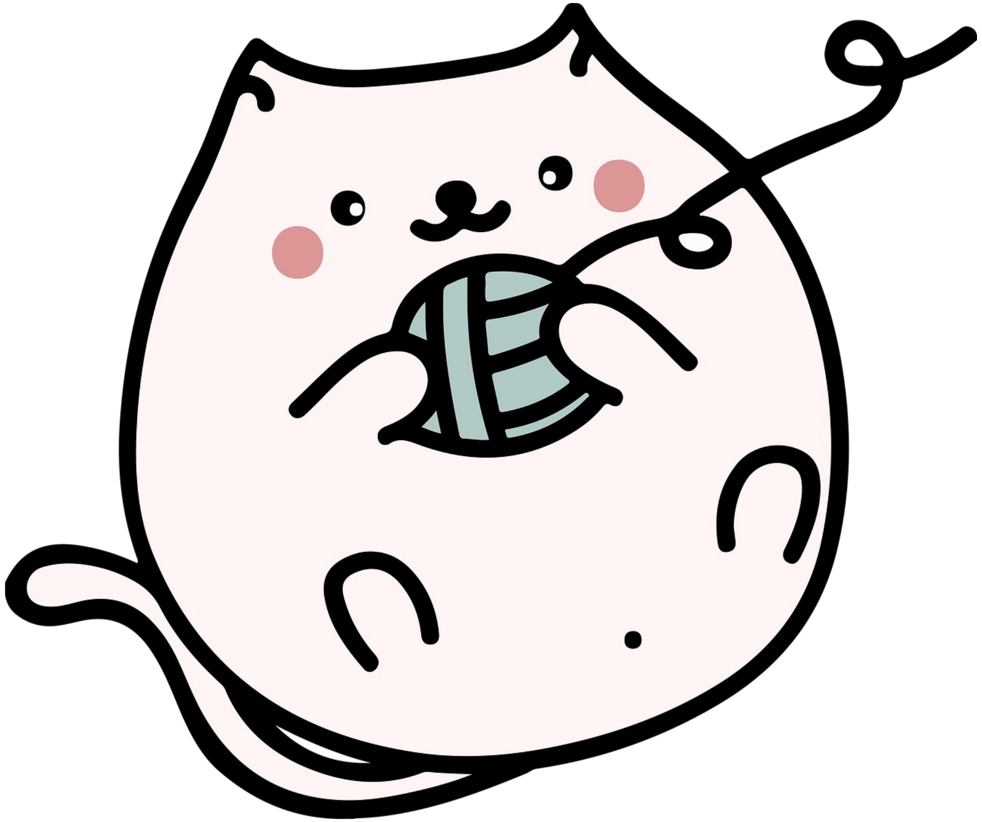
We are presented with an alternative view.
Not all cats are meanies.

We review the evidence.
I travel the entire universe and have a meet and greet with every single cat. About 78 percent of our encounters were very pleasant.

We determine if the evidence is enough.
Are these meet and greets sufficient for capturing the overall personality of cats?

If the evidence is sufficient, we reject our preconceived notion and accept the alternative view as our new reality.
I have sufficient evidence to reject the notion that all cats are meanies and now accept that not all cats are meanies.

Continue…
Maybe not all of them are meanies, but could it be that they are all spies for an alien civilization…



We have a preconceived notion about the world.

We have a null hypothesis.

We are presented with an alternative view.

We would like to investigate an alternative hypothesis.

We review the evidence.

We do inferential statistics (calculate the test statistic) using our sample data.

We determine if the evidence is enough.

Is our p value lower than the significance level?

If the evidence is sufficient, we reject our preconceived notion and accept the alternative view as our new reality.

If so, we reject our null hypothesis and conclude that the alternative hypothesis is “true.”

Continue…

Scientific research continues…

All hypothesis tests are pass/fail!

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